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Sequential Cdk1 and Plk1 phosphorylation of caspase-8 triggers apoptotic cell death during mitosis

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ABSTRACT

Caspase-8 is crucial for cell death induction, especially via the death receptor pathway. The dysregulated expression or function of caspase-8 can promote tumor formation, progression and treatment resistance in different human cancers. Here, we show procaspase-8 is regulated during the cell cycle through the concerted inhibitory action of Cdk1/cyclin B1 and polo-like kinase 1 (Plk1). By phosphorylating S387 in procaspase-8 Cdk1/cyclin B1 generates a phospho-epitope for the binding of the PBD of Plk1. Subsequently, S305 in procaspase-8 is phosphorylated by Plk1 during mitosis. Using an RNAi-based strategy we could demonstrate that the extrinsic cell death is increased upon Fas-stimulation when endogenous caspase-8 is replaced by a mutant (S305A) mimicking the non-phosphorylated form. Together, our data show that sequential phosphorylation by Cdk1/cyclin B1 and Plk1 decreases the sensitivity of cells toward stimuli of the extrinsic pathway during mitosis. Thus, the clinical Plk1 inhibitor BI 2536 decreases the threshold of different cancer cell types toward Fas-induced cell death.

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1. Introduction

Cell cycle checkpoints control the order and timing of cell cycle progression to warrant that critical mechanisms such as DNA replication and chromosome segregation are completed with high accuracy. Checkpoint activation results in cell cycle arrest to provide time for repair. Loss of checkpoint control leads to genomic instability and contributes to the transformation of normal cells into cancer cells. However, to permit repair and prevent unnecessary tissue damage, it might be required to limit apoptosis in response to low levels of damage

and during checkpoint arrest. Thus, a fine tuning between cell cycle control and the threshold of apoptosis initiation is required for an appropriate response to cellular damage during cell cycle progression.

Two intensively studied cell cycle kinases are polo-like kinase 1 (Plk1) and cyclin-dependent kinase 1 (Cdk1). Plk1 encompasses two functional domains: the N-terminal kinase domain and the C-terminal polo-box domain (PBD). The PBD mediates the recruitment of the catalytic activity of various Plks to specific subcellular locations (Jang et al., 2002; Lee et al., 1998). According to the current model Plk1 is targeted

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to specific subcellular locations or substrates through binding of the PBD to a protein that has been phosphorylated by a Pro-directed kinase, a mechanism, which is considered as ‘non-self-priming’ process (Lee et al., 2008). Plk1 and Cdk1/cyclin B1 coregulate nuclear envelope breakdown, centrosome separation, spindle assembly, chromosome condensation and Golgi fragmentation, respectively (Nigg, 2001). Moreover, many studies provided convincing evidence that Cdk1 and Plk1, which are overexpressed and have enhanced kinase activity in many tumor types, are potential targets for cancer therapy (McInnes and Wyatt, 2011; Rizzolio et al., 2010; Strebhardt, 2010).

Initiation of apoptosis is triggered by different stimuli, including UV or γ -irradiation, chemotherapeutic drugs, and death receptor signaling (Krammer et al., 2007). Interaction of the death receptor CD95 with its natural ligand, Fas ligand (CD95L) (Suda et al., 1993), or with agonistic antibodies induces apoptosis (Trauth et al., 1989; Yonehara et al., 1989). The death-inducing signaling complex (DISC), which encompasses oligomerized CD95 receptors, the adaptor molecule FADD, procaspase-8, procaspase-10, and c-FLIP_{L/S/R}, is formed immediately upon CD95 stimulation (Golks et al., 2005; Kischkel et al., 1995; Muzio et al., 1996; Oberst and Green, 2011; Sprick et al., 2000; Walczak and Sprick, 2001). Homotypic interactions between individual components of the DISC support the correct functional organization of the multimeric complex. The initiator of the extrinsic pathway, procaspase-8, contains two tandem death effector domains (DEDs) followed by a large p18 and a small p10 catalytic subunit. Following the binding of the ligand to its receptor the DISC assembles and according to the induced proximity model, in which high local concentrations and an appropriate mutual orientation of procaspase-8 are required, it is proteolytically processed and activated (Boatright et al., 2003; Muzio et al., 1998). A sequential two-step-process characterizes the autocatalytic cleavage of procaspase-8: (1) the first cleavage occurs at D374 and produces two subunits p43/p41 and p12. (2) The second step involves cleavage of D216 and D384 leads to the generation of the two active subunits p10 and p18 and the pro-domain p26/p24. The processed caspase-8 heterotetramer (p18₂-p10₂) is released into the cytoplasm, where it activates executioner caspases that in turn cleave death substrates leading to a characteristic apoptotic phenotype (Fischer et al., 2006).

Here, we provide the first demonstration that the PBD of Plk1 binds to a phospho-epitope within procaspase-8 that is generated by Cdk1/cyclin B1. Subsequent phosphorylation at the serine residue 305 interferes with the activity of caspase-8 and inhibits its processing. A small molecule inhibitor of Plk1, BI 2536, sensitizes mitotic cells to Fas-mediated apoptosis.

2. Methods

2.1. Antibodies

Commercially available antibodies against Tubulin (clone B-5-1-2 # T6074); β -actin (clone AC-15 #A5441), Vinculin

(clone VIN-11-5 #V4505), Anti-Flag (FlagM2 #F1804; all from Sigma), Plk1 (F-8 #sc-17783), Cdk1 (17, #sc-54), Cyclin B1 (GNS1 #sc-245), GFP (B-2 #9996), GST (B-14 #sc138; all from Santa Cruz), Plk1 (clone 35-206 #05-844), Plk1 (NT #06-813), phospho-histone H3 S10 (clone 3H10 #05-806; all from Millipore), Survivin (#ADI-AAP-275-E), caspase-8 (C15 # ALX-804-429-C100; both Enzo), Aurora-B (AIM1 #3094), cleaved caspase-8 (18C8 #9496), Caspase-3 (3G2 #9668) and PARP (46D11 #9532; all from Cell Signaling Technology) were used according to the manufacturer's recommendations. To generate a polyclonal antibody against caspase-8 phosphorylated on S305 rabbits were immunized by Eurogentec using the peptide IYQLMDHpSNMDCF followed by affinity-purification. The phospho-specific antibody against Casp8-pS387 is described elsewhere (Matthess et al., 2010). Secondary horseradish peroxidase-linked antibodies were from Amersham. Secondary antibodies for indirect immunofluorescence were goat anti-rabbit IgG and goat anti-mouse IgG coupled to Alexa Fluor 488 (Green), Alexa Fluor 594 (Red) or Alexa Fluor 647 (Far-Red) were from Invitrogen.

2.2. Proximity ligation assay (PLA)

PLA detection of the caspase-8-Plk1 interaction was performed using Duolink in situ PLA kit (Olink Bioscience) according to manufacturer's instructions.

2.3. Cell culture, transfection and synchronization

Cell lines were purchased from the American Type Culture Collection and grown in the recommended medium supplemented with 10% heat-inactivated FCS at 37 °C in a humidified environment with 5% CO₂.

To generate knockdown clones for caspase-8, cells were transfected with plasmids in which the H1 promoter drives the expression of an shRNA targeting caspase-8 (5'-TCACAGACTTTGGACAAAG -3'). Stably transfected cells were selected for 4 weeks using 1.5 mg/ml G418 (PAA) and isolated for single clones. Transfections were performed by using TurboFect (Thermo Scientific) according to the manufacturer's protocol. Synchronizations were performed as described (Raab et al., 2011). Briefly, for mitotic shake-off, cells were incubated for 16 h with nocodazole (50 ng/ μ l), shaken-off and washed with PBS before released into media.

2.4. Western blotting

Whole-cell lysates were made in modified RIPA lysis buffer (50 mM Tris-HCl at pH 7.5, 150 mM NaCl, 1% NP-40, 0.5% sodium deoxycholate, 0.1% SDS, 10% Glycerol, 50 mM NaF and complete protease and phosphatase cocktail, Roche) for 10 min on ice, and boiled in 2 $\times$ Laemmli loading buffer. Protein samples were resolved by SDS-PAGE and transferred onto PVDF membrane using TransBlot Turbo (BioRad), which was blocked in 5% BSA in TBST and probed with the indicated antibodies.

2.5. Generation of plasmids and mutagenesis

Caspase-8 was inserted into a pGEX5x-3 (GE Healthcare), into a 3xFlag-tagged and EGFP-tagged pcDNA3.1-Hygro (Invitrogen) vector. Subdomains of caspase-8 were HindIII/XbaI inserted into pGEX and pcDNA3.1-Hygro vectors. The 3xFlag-tagged caspase-8 sequence was inserted into the pcDNA3.1-V5 vector (Invitrogen). Deletion clones were generated by standard cloning techniques. Mutants were generated by site-directed mutagenesis using the Quick Change protocol and Pfu Ultra II Fusion HS DNA Polymerase (Stratagene). All constructs were confirmed by sequencing.

2.6. Immunoprecipitation and phospho amino acid analysis

Both methods were performed as described previously (Matthess et al., 2010). For immunoprecipitation Protein G Sepharose 4 Fast Flow (Amersham) was used.

2.7. GST pulldown

The expression of recombinant GST-proteins was induced in *Escherichia coli* BL21 cells at 37 °C for 2 h by the addition of 1 mM IPTG. GST-fused proteins were purified with the Cell Lytic B protocol (Sigma #B7435) first and then incubated with lysates of HeLa cells transfected with the Flag-Cdk1 expression vector in TBSN buffer (20 mM Tris, pH 8.0, 150 mM NaCl, 0.5% Nonidet P-40, 5 mM EGTA, 0.5 mM Na₃VO₄, 20 mM *p*-nitrophenyl phosphate) supplemented with protease inhibitors (Complete Mini EDTA-free Protease Inhibitor Cocktail, Roche Diagnostics) at 4 °C for 2 h. GST, GST-caspase-8 and GST-fused subdomains of caspase-8 were adsorbed to glutathione-Sepharose 4B (GE Healthcare) for an additional 1 h. The bound proteins were resolved by SDS-PAGE and transferred to an Immobilon-P membrane (Millipore). The blots were then probed with anti-Flag antibodies to detect the Flag-Plk1 protein.

2.8. Cell cycle analysis

Cell cycle analysis was carried out as described previously (Yuan et al., 2006). Briefly, cells were washed with PBS, treated with RNase A, and stained with PI (Propidium Iodide). Flow cytometry was performed using a FACScan and Cell Quest software (Becton Dickinson). The determination of the mitotic indices and mitotic phase distribution was performed as previously described (Reindl et al., 2008). All experiments were performed in triplicate.

2.9. Immunofluorescence assays

For immunofluorescence analysis, HeLa cells were grown on a glass coverslip, fixed with 4% paraformaldehyde for 15 min, washed with PBS and 10 min permeabilized with 0.2% Triton-X before addition of the appropriate primary and secondary antibodies. Microscopy was performed with a Zeiss Axio Imager.Z1 microscope equipped with a 40× objective, and the images were captured and processed using AxioVision Software (Zeiss).

2.10. In vitro kinase, caspase-3/7 and apoptosis assays

Immunoprecipitations using Plk1- or Cdk1-antibodies to measure the kinase activities were performed as described (Raab et al., 2011). *In vitro* kinase assays were performed using 10× buffer (New England Biolabs) supplemented with 0.05 mM ATP and 1 μCi of [γ -³²P]ATP (3000 Ci/mmol, Amersham Pharmacia) for 30 min at 30 °C in the presence of bacterially expressed and purified GST-caspase-8 fusion proteins. Samples were resolved by SDS/PAGE and subjected to autoradiography. Caspase-3/7 activity was determined using the Caspase-Glo 3/7 Assay kit (Promega) according to the manufacturer's instructions. Cells were processed using a Vybrant Apoptosis Assay Kit#2 (Alexa Fluor 488 annexin V/propidium iodide staining) according to the manufacturer's protocol (Invitrogen) and analyzed by flow cytometry using a FACScan (Becton Dickinson).

2.11. Statistical methods

Experimental *in vitro* data are presented as mean ± standard deviations from three or more independent experiments. Two-way analysis of variance (ANOVA) (GraphPad Prism; GraphPad Software, Inc., San Diego, CA) was done to consider random effects of individual gels and different treatments. For two-way ANOVAs, all treatment groups were compared with control cells.

3. Results

3.1. Cdk1/cyclin B1 generates a binding site for Plk1 in procaspase-8

To elucidate the cellular functions of procaspase-8 in detail, we incubated GST-procaspase-8 with lysates from mitotically active cancer cells and subjected the obtained pulldown complexes to mass spectrometry (data not shown) followed by matrix-assisted laser desorption ionization-time of flight analysis. This investigation identified several procaspase-8 interacting partners including Plk1.

To determine the cellular relevance of our finding, we analyzed the association between procaspase-8 and Plk1 in HeLa cells. Following the immunoprecipitation of cell extracts using caspase-8- or Plk1-specific antibodies complexes containing procaspase-8 and Plk1 were detected in nocodazole-treated cells (Figure 1A and B), but not in asynchronously growing HeLa cells (Figure 1A). Cdk1 and cyclin B1 were also components of this immunocomplex (Figure 1A) confirming our previous results (Matthess et al., 2010). Co-expression of GFP-Plk1 and Flag-tagged caspase-8 followed by Flag-IP and western blotting using GFP-specific antibodies provided further evidence for the interaction of both proteins (Figure 1C). Using the Proximity Ligation Assay (PLA) we confirmed the close proximity of both proteins in mitotic, but not in non-mitotic cells (Figure 1D).

Next, using lysates of HEK 293T cells transfected with expression constructs for Flag-Plk1 or Flag-Plk1-PBD, we performed pulldown assays with GST-fused subdomains of procaspase-8 to determine the regions involved in the

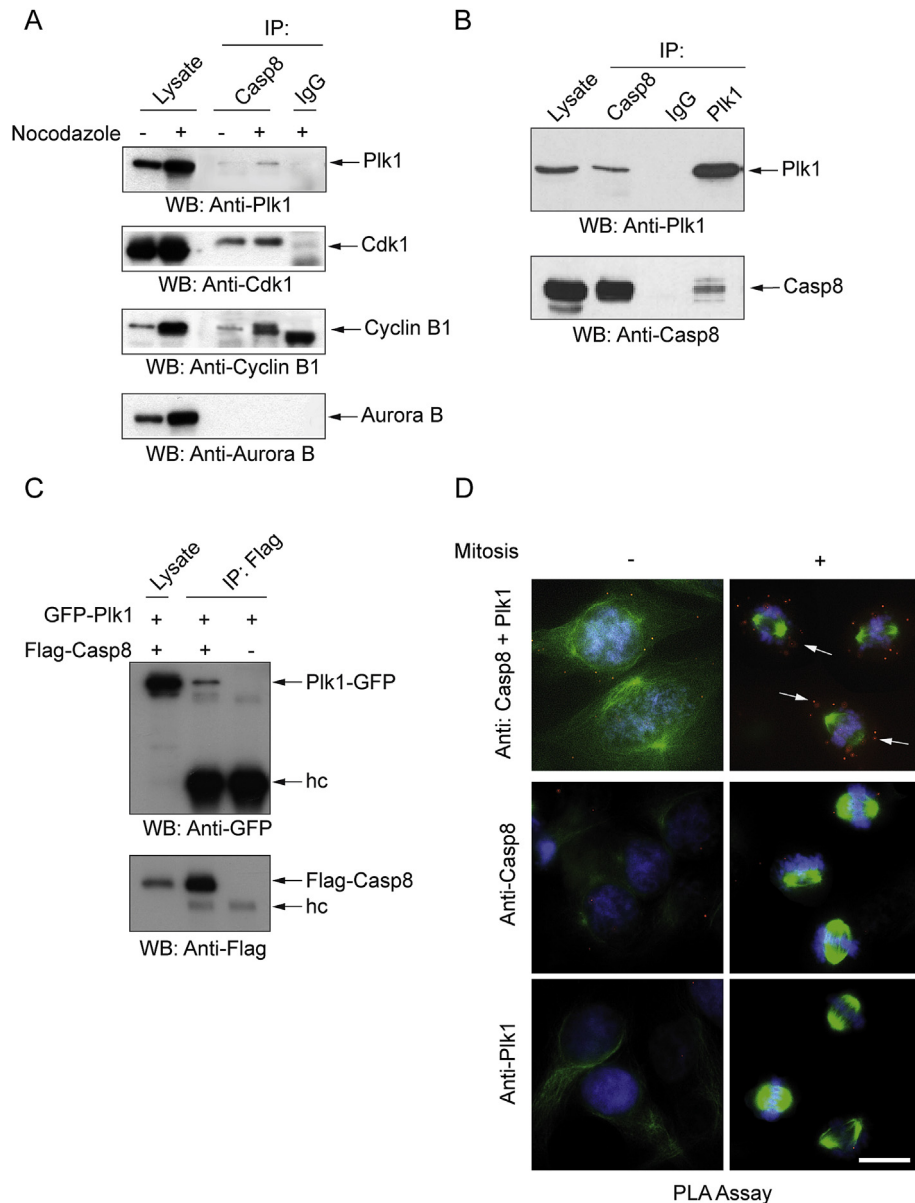


Figure 1 – Procasase-8 interacts with Plk1 in mitotic cells. (A) Lysates from asynchronous (–) or nocodazole-treated (+) HeLa cells were immunoprecipitated with caspase-8-antibodies. Anti-mouse IgG was used as a control. The immunoprecipitated proteins were resolved by SDS-PAGE and immunoblotted for Plk1, Cdk1, cyclin B1 and Aurora-B. Lanes 1 and 2 show 2.5% total input. (B) Lysates from mitotic HeLa cells were immunoprecipitated with caspase-8 or Plk1-specific antibodies. Anti-mouse IgG was used as a control. The immunoprecipitated proteins were resolved by SDS-PAGE and immunoblotted for Plk1 or caspase-8. Lanes 1 and 2 show 2.5% total input. (C) Lysates of HeLa cells that had been transfected with a Flag epitope-tagged caspase-8 (Flag-Casp8) and a GFP-fused Plk1 expression construct were immunoprecipitated with Flag-specific antibodies. The immunoprecipitated proteins were resolved by SDS-PAGE and immunoblotted for GFP and Flag. Lane 1 shows 2.5% total input. (D) *In situ* PLA detection of caspase-8 and Plk1 in nocodazole-treated (+) and exponentially growing cells (–). Red dots depicted by arrows indicate interactions between caspase-8 and Plk1. Scale bar, 10 μ m.

association of both proteins *in vitro*. While low levels of Flag-Plk1 or Flag-Plk1-PBD associated with the N-terminal portion of caspase-8 comprising the two DED domains (GST-NT), the C-terminal part of caspase-8 containing the large and the small catalytic subunit (GST-CT) showed strong signals (Figure 2A, left and right panels). The expression of GST-fused proteins was controlled by western blotting (Figure 2A, lower left and right panels). Using lysates of cells transfected

with expression constructs for GFP-procaspase-8 for a pull-down assay with GST-fused Plk1-PBD, we revealed that both, endogenous and GST-fused procaspase-8, interacts with the PBD of Plk1 (Figure 2B). Previously, it could be demonstrated that the PBD binds to a phospho-epitope (Elia et al., 2003a), indicating that a priming phosphorylation is an important step to promote the PBD-mediated interaction. Thus, we searched the sequence of procaspase-8 for the core consensus

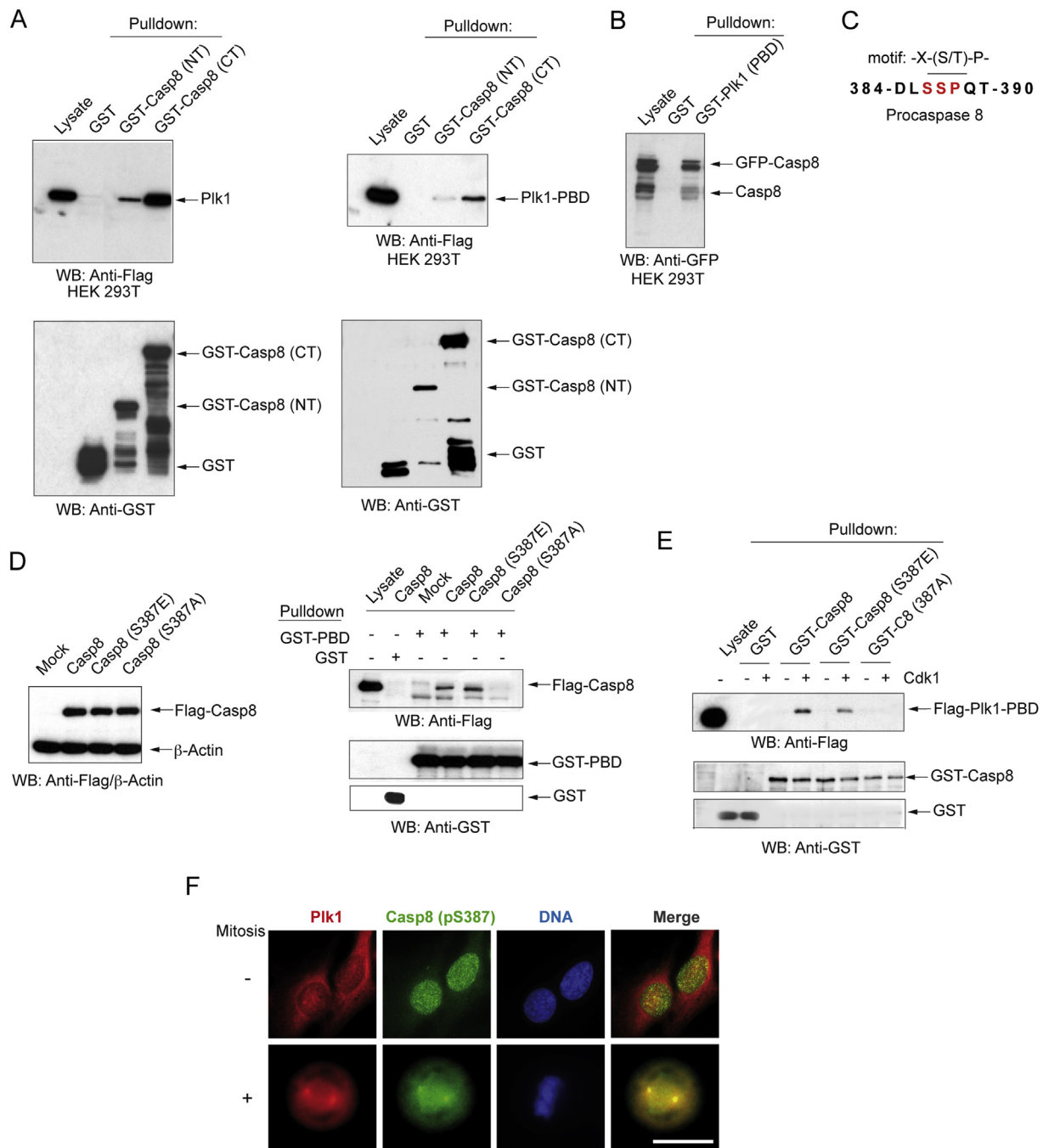


Figure 2 – Phosphorylation of S387 stimulates the interaction of procaspase-8 and Plk1. (A) Bacterially expressed, purified GST-fused N-terminal (GST-NT) and C-terminal (GST-CT) subdomains of procaspase-8 were incubated with lysates of HEK 293T cells that had been transfected with a Flag epitope-tagged Plk1 or a Flag epitope-tagged Plk1-PBD expression construct for pull-down assays. The Flag-Plk1 or Flag-Plk1-PBD protein that associated with GST-fused subdomains of caspase-8 was detected by immunoblot with an anti-Flag antibody. (B) Bacterially expressed, purified GST-fused Plk1-PBD was incubated with lysates of HEK 293T cells that had been transfected with a GFP-tagged caspase-8 expression construct for pull-down assays. The GFP-caspase-8 protein that associated with GST-fused Plk1 was detected by immunoblot with an anti-GFP antibody. (C) The putative PBD binding motif in procaspase-8 is depicted. (D) Bacterially expressed, purified GST-fused Plk1-PBD or GST alone were incubated with lysates of HEK 293T cells that had been transfected with expression constructs for Flag-tagged caspase-8 or caspase-8 mutants (S387A, S387E) for pull-down assays. The Flag-tagged caspase-8 protein that associated with GST-fused PBD was detected by immunoblot with an anti-Flag antibody. (E) Bacterially expressed, purified GST-fused caspase-8 or caspase-8 mutants (S387A, S387E) were first pre-incubated with Cdk1/cyclin B1 in a cold kinase assay for subsequent incubation with lysates of HEK 293T cells that had been transfected with expression constructs for Flag-tagged PBD for pull-down assays. The Flag-tagged Plk1-PBD that associated with GST-fused caspase-8 was detected by immunoblot with an anti-Flag antibody. (F) Untreated HeLa cells (top) or treated with nocodazole for 12 h. Cells were fixed and costained with antibodies for Plk1 and caspase-8-pS387. DNA was stained with DAPI (4',6-diamidino-2-phenylindole). Scale bar, 10 μ m.

motif S-pT/pS-P-X that functions as recognition sequence for Plk1-PBD binding. We found phospho-S387 to be embedded in the S-pS-P-Q motif (Figure 2C) that fulfills the criteria for PBD binding and is a Cdk1/cyclin B1 phosphorylation site (Matthess et al., 2010). To examine whether binding of Plk1 requires the phosphorylation of S387, we transfected cells with expression constructs for Flag-procaspase-8 wild-type (WT) or corresponding single mutants of procaspase-8 (S387A or S387E). Lysates of transfected cells (Figure 2D, left panel) were subjected to pulldown assays with GST-fused Plk1-PBD or GST alone. While procaspase-8 (WT) and its phosphomimicking form (S387E) were both able to associate with the Plk1-PBD, the non-phosphorylatable mutant (S387A) could not bind (Figure 2D, right panel). Next, we subjected GST-fused caspase-8 to a cold *in vitro* kinase assays with Cdk1/cyclin B1 and performed a pulldown in Flag-Plk1-PBD expressing lysates. In contrast to the non-phosphorylatable caspase-8 (S387A), both, the phosphomimicking form (S387E) and caspase-8 (WT) phosphorylated by Cdk1/cyclin B1, could bind with a strongest signal for the phosphorylated form indicating that phosphorylation of S387 stimulates the PBD-mediated interaction of Plk1 and caspase-8 (Figure 2E). Moreover, to determine the subcellular localization of the interaction, we used caspase-8 phospho-S387 antibodies and monoclonal Plk1-antibodies and detected colocalization of Plk1 and caspase-8(pS387) at centrosomes in metaphase cells, but not in interphase cells (Figure 2F).

3.2. Plk1 phosphorylates procaspase-8 at S305

To determine whether procaspase-8 is an authentic substrate of Plk1, we incubated purified recombinant GST-fused subdomains of procaspase-8 with Plk1 in the presence of [γ - 32 P]ATP (Figure 3A). Only the C-terminal domain (CT), containing p18 and p10, was labeled (Figure 3A). The amino acid analysis demonstrated that the site(s) phosphorylated by Plk1 was mainly serine residues (Figure 3A). To further narrow down the target region, we analyzed GST-fused p18 and p10 in an *in vitro* kinase assay. While Cdk1/cyclin B1 was able to phosphorylate p10, Plk1 phosphorylated p18 (Figure 3B). Next, we substituted alanine for serine at all serine sites within p18. Figure 3C, which depicts only a small collection of mutants tested in kinase assays with Plk1, revealed that only p18-S305A was not phosphorylated by Plk1, indicating that the sequence including S305, which is in good agreement with the Plk1 consensus motif (D/E-X-S/T- Φ ; X, any amino acid; Φ , a hydrophobic amino acid) (Nakajima et al., 2003) is a Plk1 phosphorylation site *in vitro*. Further kinase assays using the wild-type or the mutated C-terminal domain (CT) including p18 and p10 (S305A or S387A) as substrate confirmed that while the mutation of S387A completely abolished the phosphorylation by Cdk1/cyclin B1, S305A blocked the phosphorylation by Plk1 (Figure 3D). To further determine whether procaspase-8 is a *bona fide* substrate of Plk1, we examined the phosphorylation of the full-length protein. Site-directed mutagenesis of S387, which is in the PBD binding site for Plk1 did not block the phosphorylation of full-length procaspase-8 completely but reduced the phosphorylation signal, suggesting that S387 is a critical determinant for the phosphorylation of full-length procaspase-8 by Plk1 (Figure 3E).

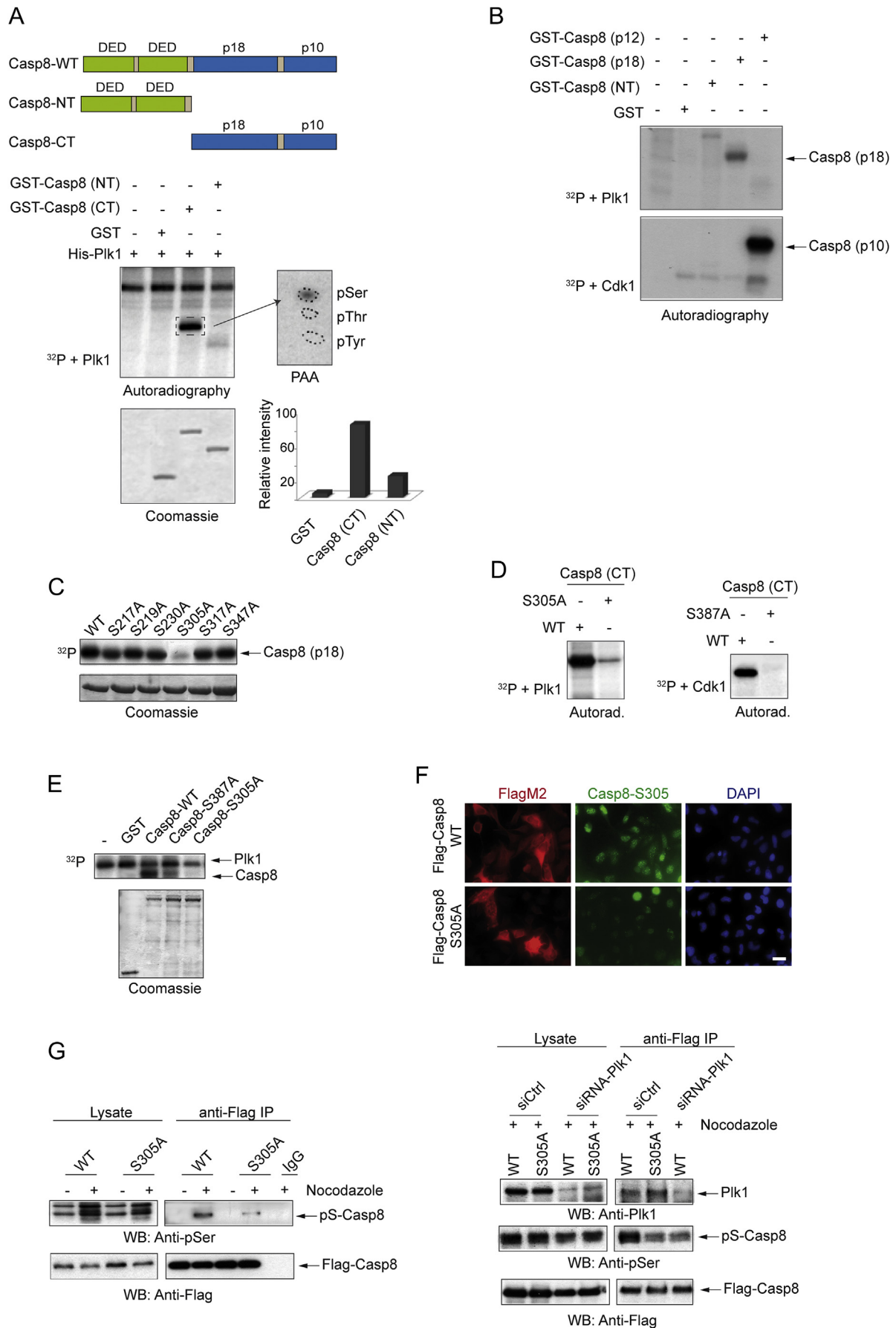
Site-directed mutagenesis of S305 blocked the phosphorylation completely providing further evidence for S305 as a major phosphorylation site for Plk1 of full-length procaspase-8 *in vitro* (Figure 3E).

We generated a phospho-antibody using the peptide IYQLMDHpSNMDCF for the immunization of rabbits to explore the significance of S305 phosphorylation *in vivo*. While the new antibody did not recognize the denatured caspase-8 protein in western blot experiments, the signal of phospho-caspase-8 detected by immunofluorescence microscopy was significantly reduced in caspase-8-depleted cells that were transfected with Flag-caspase-8 (S305A) compared to caspase-8 (WT) transfected cells (Figure 3F).

To further explore caspase-8 phosphorylation *in vivo*, cells were transfected with Flag-tagged forms of caspase-8 (WT or S305A) and incubated with nocodazole to enrich mitotic cells with high Plk1 activity. Cell extracts were subjected to anti-Flag IP, and analyzed by immunoblot to detect the phosphorylation of caspase-8 by using a phosphoserine antibody as described (Yuan et al., 2007). The phosphorylation of caspase-8 (WT) at serine was clearly increased in mitotic cells (Figure 3G, left panel). The S305A mutation significantly reduced caspase-8 phosphorylation at serine, indicating that S305 is a major serine phosphorylation site in mitotic cells *in vivo* (Figure 3G, left panel). To test whether procaspase-8 could be phosphorylated by Plk1 *in vivo*, cells treated with mismatch (MM) siRNA or siRNA-Plk1 were transfected with Flag-tagged caspase-8 (WT or S305) and incubated with nocodazole (14 h). The analysis of the anti-Flag IP with a phosphoserine antibody showed that silencing of endogenous Plk1 reduced the phosphorylation of procaspase-8 (WT) at serine, compared with phosphorylated procaspase-8 (WT) detected in control cells, suggesting that Plk1 phosphorylates caspase-8 *in vivo* (Figure 3G, right panel). The S305A mutation also significantly reduced procaspase-8 phosphorylation in cells with high Plk1 activity, indicating that S305 of procaspase-8 is the Plk1 phosphorylation site *in vivo* (Figure 3G, right panel).

3.3. The phosphorylation of procaspase-8 by Plk1 inhibits its processing upon Fas-stimulation

Remarkably, the phosphorylation site S305 for Plk1 is located close to D302, which represents a site in procaspase-8 for low-penetrance mutations in breast cancer (Cox et al., 2007). The caspase-8 D302H mutation belongs to a group of nine SNPs that show an association with breast cancer indicating that an amino acid exchange at this position is functional relevant and contributes to breast cancer development. To evaluate whether phosphorylation of this region by Plk1 can affect the activity of procaspase-8 upon Fas-stimulation, we enriched cells in mitosis by nocodazole treatment as measured by FACS (Figure 4A, left panels) associated with a strong accumulation of Plk1, cyclin B1 and phospho-histone H3 as shown by western blotting (Figure 4A, middle panel) and subsequently added the small molecule Plk1 inhibitor BI 2536 *in vivo* (Steegmaier et al., 2007). Kinase assays with immunoprecipitated Plk1 using monoclonal antibodies showed high Plk1 activity in cells treated with nocodazole alone and almost completely blocked Plk1 activity in



nocodazole- and BI 2536-treated cells (Figure 4A, right panels). To examine the correlation of procaspase-8 processing (Figure 4B, left panel) and Plk1 activity in nocodazole-treated cells, we analyzed protein extracts of HeLa cells treated with FasL in the absence or presence of BI 2536. Fas-stimulation led to increased levels of p18 in BI 2536-treated mitotic cells compared to mitotic control cells suggesting that inhibition of Plk1 stimulates FasL-induced processing of procaspase-8 (Figure 4B, right panel). While small molecule ATP-competitive Plk1 inhibitors may target multiple protein kinases, carefully selected siRNAs are presumed to have exquisite target specificity. Thus, we examined procaspase-8 processing in mitotic, Plk1-depleted ovarian cancer cells (SKOV-3). Fas-stimulation led to increased levels of p18 in mitotic cells with downregulated Plk1 compared to siRNA-treated mitotic control cells confirming that Plk1 contributes to an inhibition of procaspase-8 in mitotic cells (Figure 4C, left panel). To corroborate this correlation in various living cells, we tested cancer cell lines of different origin (ovarian (SKOV-3), lung (A549), cervical (HeLa)) enriched in mitosis treated with siRNA-Plk1 or BI 2536 (Figure S1). Treatment of cancer cells with BI 2536 (Figure S1 A, B, left panels) or Plk1-specific siRNA (Figure S1 C, D left panel) led to increased levels of p18 and apoptosis (Figure S1 C, D, right panels) upon Fas-stimulation compared to untreated mitotic cells indicating that Plk1 activity interferes with the activation of the extrinsic pathway i.e. caspase-8 processing in different cancer cell types.

3.4. In cells (type II) with a compromised intrinsic death pathway inhibition of Plk1 sensitizes to Fas-mediated apoptosis

Furthermore, to evaluate a putative contribution of the intrinsic death pathway to the apoptotic response that we observed upon Fas-stimulation, we analyzed HCT116 cells enriched in mitosis with a disrupted mitochondrial death pathway due to a lack of a functional BAX gene (Zhang et al., 2000). Inhibition of Plk1 activity using BI 2536 led to a robust

increase of procaspase-8 processing as measured by monitoring p18 levels compared to untreated cells supporting our hypothesis that Plk1 activity blocks the extrinsic death pathway (Figure S2).

3.5. Analysis of the extrinsic cell death under conditions of modulated Plk1 activity or using mutated forms of procaspase-8 in living cells

Next, we evaluated whether phosphorylation of S305 controls Fas-induced processing of procaspase-8 in living cells. To this end, we overexpressed procaspase-8 (WT) and its mutants (non-phosphorylatable procaspase-8 S305A, phosphomimicking procaspase-8 S305E) in HeLa cells followed by Fas-stimulation. Procaspase-8 S305E seemed to have the highest stability in this experiment during an observation period of 6 h (Figure 5A, left panel). Still, this result might be flawed by the activity of endogenous caspase-8 activity. Thus, we performed a rescue experiment and depleted endogenous caspase-8 by stable expression of shRNA targeting the untranslated region of caspase-8 mRNA and replaced it by similar amounts of transfected procaspase-8 (WT) or its mutants (S305A, S305E) (Figure 5A, right panel). The replacement of endogenous procaspase-8 with the S305A mutant or procaspase-8 (WT) followed by Fas-stimulation led to pronounced caspase-8 processing in exponentially growing cells (Figure 5A, right panel). In contrast, the expression of the S305E mutant in a caspase-8-depleted HeLa cell population, protected against Fas-mediated apoptosis (Figure 5A, right panel) indicating the inhibitory role of S305 phosphorylation for the processing of procaspase-8.

Next, we correlated the activities of Plk1 and procaspase-8 in the context of cell death. To address this aspect, we transfected HeLa cells with wild-type Plk1 or its activity mutants (K82M, T210D) and examined the extrinsic death pathway. Transient transfections of HeLa cells with wild-type Plk1 followed by Fas-stimulation showed reduced levels of procaspase-8 processing compared to mock-transfected cells (Figure 5B). Moreover, we analyzed HeLa cells stably

Figure 3 – Mapping of Plk1 phosphorylation sites in procaspase-8. (A) Scheme of procaspase-8 and its subdomains (left panel). Bacterially expressed, purified GST-fused N-terminal (GST-NT) and C-terminal (GST-CT) subdomains of procaspase-8 were incubated with Plk1 and [γ - 32 P]ATP, analyzed by SDS-PAGE, and visualized by autoradiography (left panel). This was followed by phospho amino acid analysis (right panel). Recombinant GST-fused subdomains of procaspase-8 were analyzed by SDS-PAGE (lower left panel). Relative radioactive signal intensity of GST-fused N-terminal (GST-NT) and C-terminal (GST-CT) subdomains of caspase-8 phosphorylated by Plk1 is depicted as bar graph (lower right panel). (B) Bacterially expressed, purified GST-fused N-terminal (GST-NT) and C-terminal (GST-CT) subdomains of procaspase-8 were incubated with Plk1 or Cdk1/cyclin B1 and [γ - 32 P]ATP, analyzed by SDS-PAGE, and visualized by autoradiography. (C) Wild-type (WT) and mutated (S217A, S219A, S230A, S305A, S317A, S347A) forms of the p18 subdomain of caspase-8 were subjected to kinase assays with Plk1 including [γ - 32 P]ATP, analyzed by SDS-PAGE, and autoradiography. (D) Comparative phosphorylation of the mutated C-terminal domain (CT) of procaspase-8 including p18 and p10 (S305A, S387A) in kinase assays including [γ - 32 P]ATP using Plk1 or Cdk1/cyclin B1, analyzed by SDS-PAGE, and autoradiography. (E) Wild-type (WT) and mutated (S387A, S305A) recombinant full-length GST-caspase-8 proteins were subjected to kinase assays using Plk1 including [γ - 32 P]ATP, analyzed by SDS-PAGE, and autoradiography. (F) Caspase-8-depleted HeLa cells were transfected with expression vectors for wild-type caspase-8 or caspase-8 (S305A), fixed and costained with antibodies for caspase-8 (pS305). DNA was stained with DAPI (4',6-diamidino-2-phenylindole). Scale bar, 10 μ m. (G) Cells were transfected with Flag-Casp8 WT or Flag-Casp8 (S305A) and treated with or without nocodazole. Lysates were prepared and subjected to anti-Flag antibody IP, followed by western blot analysis with an antiphospho-serine antibody. The same membrane was stripped and probed with an anti-Flag antibody (left panel). Cells were transfected with siRNActrl or siRNA-Plk1, 24 h later followed by a transfection with Flag-Casp8 WT or Flag-Casp8 (S305A) and treated with nocodazole. Lysates were prepared and subjected to anti-Flag antibody IP, followed by western blot analysis with an antiphospho-serine antibody. The same membrane was stripped and probed with anti-Flag and anti-Plk1-antibodies (right panel).

overexpressing the kinase-dead form (K82M) or the hyperactive form (T210D) of Plk1 for subsequent stimulation of the extrinsic pathway. Expression of the T210D mutant suppressed activation of procaspase-8 upon Fas-stimulation (Figure 5C, left panels) and apoptotic activity as measured by caspase-3/7 activation and annexin staining compared to K82M-expressing cells (Figure 5C, middle panels). Moreover, cellular proliferation was clearly inhibited in Fas-treated cells expressing the K82M mutant compared to T210D-expressing cells (Figure 5C, right panel) supporting the model that Plk1 activity increases the survival of cancer cells.

4. Discussion

Over time normal human cells accumulate a series of genetic mutations in order to progressively develop into a broad spectrum of different cancer types. Cancer cells are therefore vulnerable to being targeted for apoptosis by the body's defensive mechanisms and avoiding this is the essential for them to develop further (Hanahan and Weinberg, 2000, 2011). Due to the enormous significance of procaspase-8 in initiating apoptosis, the function of procaspase-8 is frequently impaired in human tumors by different mechanisms including

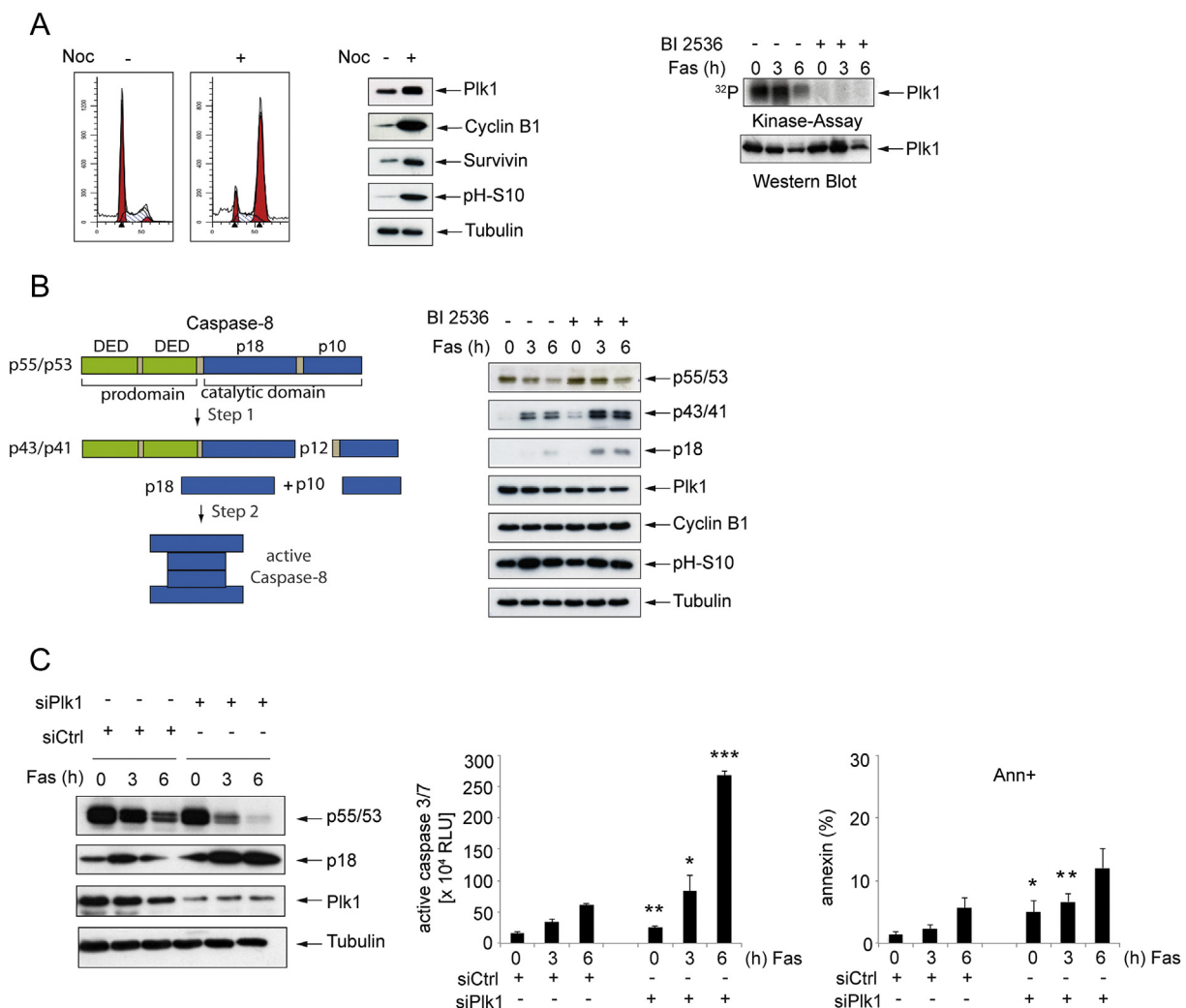


Figure 4 – The processing of procaspase-8 correlates with Plk1 activity in HeLa cells upon Fas-stimulation. (A) Untreated HeLa cells and mitotic shake-off HeLa cells treated with nocodazole were analyzed by FACS (left panel) and by western blotting (middle panel). Cells were lysed and immunoblotted for Plk1, cyclin B1, Survivin, phosphorylated histone H3 at S10 (pH-S10) and tubulin. Representative *in vitro* kinase assay using casein as substrate monitoring the activity of Plk1 immunoprecipitated from lysates of cells treated with or without BI 2536 (right panel). (B) Scheme of procaspase-8 processing and its cleavage products (left panel). Mitotic shake-off cells treated with or without 100 nM BI 2536 were stimulated with 100 ng of FasL/ml for the times indicated. Cells were lysed and immunoblotted for caspase-8, p18, Plk1, cyclin B1, pH-S10 and tubulin. (C) Mitotic shake-off cells treated with siRNA-Plk1 or a control siRNA (siCtrl) were stimulated with 100 ng of FasL/ml for the times indicated. Cells were lysed and immunoblotted for caspase-8 (p55/53), p18, Plk1, and tubulin. Caspase-3/7 activity was determined in the cell lysates using the Caspase-Glo 3/7 Assay (means $\pm$ s.d., $n = 3$, for each concentration) (middle panel). Annexin V-based measurements of cells treated with Fas were correlated with levels of apoptosis (right panel). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, Student's *t*-test, unpaired and two-tailed.

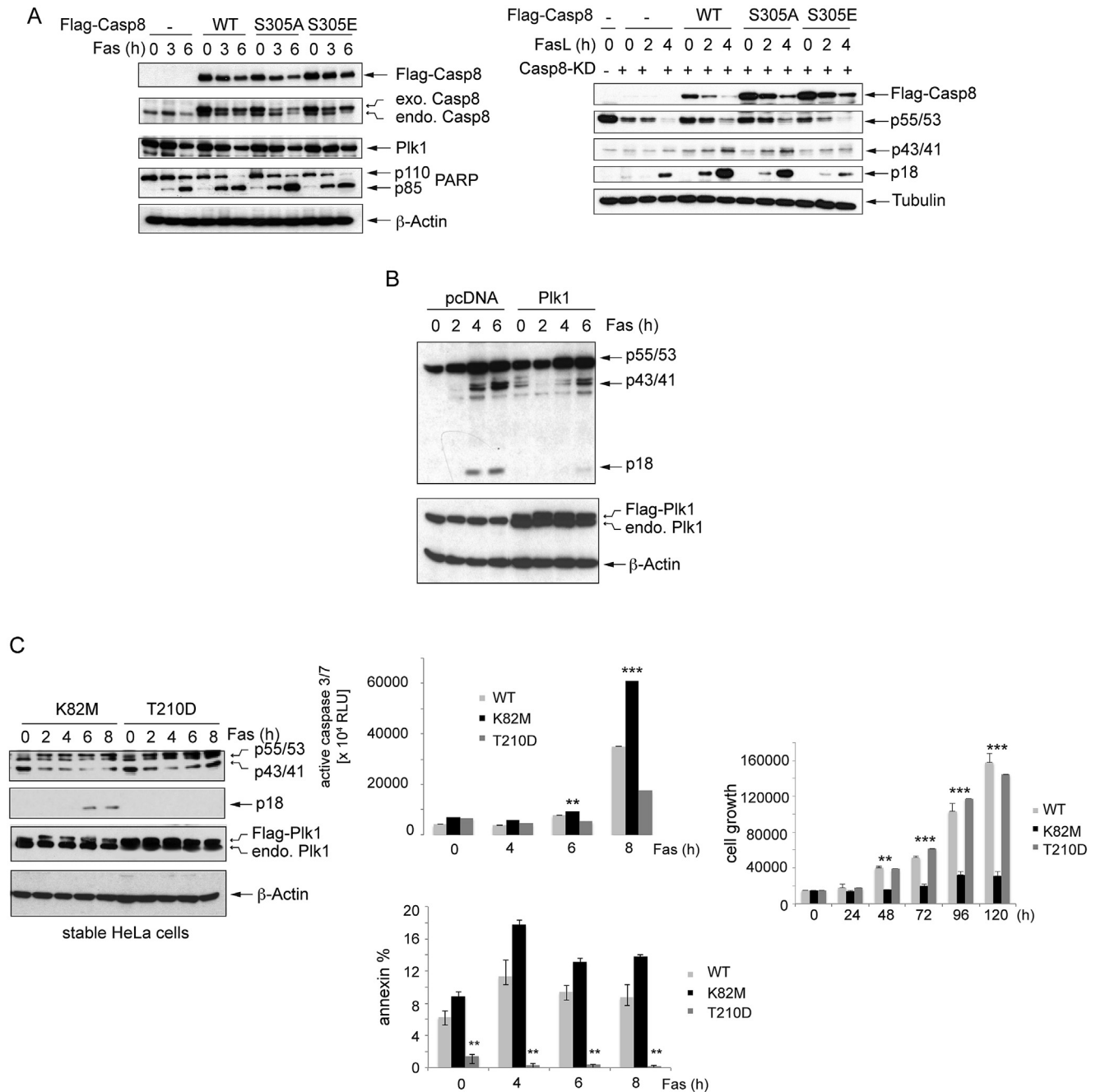


Figure 5 — Expression of a non-phosphorylatable S305 mutant of procaspase-8 or a kinase-dead form of Plk1 sensitize mitotic cells to Fas-mediated apoptosis. (A) HeLa cells were transiently transfected with vectors encoding Flag-tagged caspase-8 (Flag-WT, Flag-S305A, or Flag-S305E). At 18 h after transfection, cells were incubated with 100 ng of FasL/ml plus 1 µg of cycloheximide/ml for 3 and 6 h. Cells were lysed and immunoblotted for Flag-caspase-8, caspase-8, Plk1, PARP, and β-actin (left panel). HeLa cells stably transfected with an RNAi cassette targeting the untranslated region of caspase-8 were subsequently transfected with Flag-tagged caspase-8 (mock, WT, S305A, S305E) (right panel). On day 3, the cells were stimulated with 100 ng of FasL/ml. All experiments were performed in triplicate. Cells were lysed and immunoblotted for Flag-caspase-8, caspase-8, Plk1 and tubulin (right panel). (B) HeLa cells were transiently transfected with pcDNA or a vector encoding Flag-tagged Plk1. At 18 h after transfection, cells were incubated with 100 ng of FasL/ml plus 1 µg of cycloheximide/ml for the indicated time periods. Cells were lysed and immunoblotted for caspase-8, Plk1, and β-actin. (C) HeLa cells stably overexpressing the kinase-dead form of Plk1 (K82M) or the hyperactive form of Plk1 (T210D) were stimulated with 100 ng of FasL/ml for the indicated time periods. Cells were lysed and immunoblotted for caspase-8, Flag-Plk1, Plk1 and β-actin (left panel). Caspase-3/7 activity was determined in the cell lysates using the Caspase-Glo 3/7 Assay (means ± s.d., $n = 3$, for each concentration) (upper middle panel). Annexin V-based measurements of cells treated with FasL were correlated with levels of apoptosis (lower middle panel). Cellular proliferation of HeLa cells stably overexpressing Plk1 (K82M) or Plk1 (T210D) stimulated with 100 ng of FasL/ml was determined (right panel). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, Student's t -test, unpaired and two-tailed.

mutation, deletion, epigenetic silencing, alternative splicing and phosphorylation suggesting that caspase-8 participates in oncogenesis (Fulda, 2009). Depending on the type of cancer cells different protein kinases can phosphorylate procaspase-8 reducing its ability to undergo further processing even in the presence of death receptor-activating agents. Over the years, different groups have been able to identify several kinases that can carry out such phosphorylation. The p38 MAPK can phosphorylate procaspase-8 at S364 (Alvarado-Kristensson et al., 2004), the Src kinase at Y380 (Cursi et al., 2006), Cdk1/cyclin B1 at S387 (Matthess et al., 2010) and finally RSK2 at T263 (Peng et al., 2011).

Previous investigations have demonstrated that Cdk1/cyclin B1 has the ability to stimulate cell death by directly phosphorylating components of the cell death machinery. Cdk1/cyclin B1 phosphorylates the pro-survival Bcl-2 proteins, Bcl-2, Bcl-Xl and Mcl-1, thereby blocking their death inhibitory effects (Harley et al., 2010; Terrano et al., 2010). In addition, in cells of hematopoietic origin hyperphosphorylation of Bim at S44 by Cdk1 correlates with cell death (Mac Fhearraigh and Mc Gee, 2011). Although mitotic catastrophe induces cell death by caspase-dependent and -independent mechanisms, the Cdk1-mediated phosphorylation of caspase-2 and caspase-9 during mitotic catastrophe was shown to protect cells against apoptosis (Allan and Clarke, 2007; Andersen et al., 2009; Mansilla et al., 2006). Taken together, Cdk1 is a key player to balance pro- and anti-apoptotic signals in mitotic cells. In contrast, the role of Plk1 in apoptotic signaling is widely unexplored.

In this study, we found that Cdk1/cyclin B1 serves as priming kinase for procaspase-8 thereby promoting the interaction and subsequent phosphorylation by Plk1. The sequence context of S387 (DLSSPQT) within caspase-8 for the priming phosphorylation satisfies the optimal sequence motif recognized by the PBD, which is proposed as S-[pS/pT]-[P/X] (Elia et al., 2003b). Although phosphorylation is generally required for the target protein to bind the PBD, a recent report showed that *Drosophila* microtubule-associated protein Map205 binds strongly with the PBD of *Drosophila* Polo or human Plk1 in a phospho-independent manner (Archambault et al., 2008). Still, several physiological substrates that bind to the PBD of Plk1 in a Cdk1 phosphorylation-dependent manner have been identified, such as Cdc25C (Elia et al., 2003a; Elia et al., 2003b), the peripheral Golgi protein Nir2 (Litvak et al., 2004), and the human histone acetyltransferase binding to Orc (Hbo1) (Wu and Liu, 2008). We provide another example that this type of interaction is physiologically relevant by demonstrating that the PBD of Plk1 binds to procaspase-8 in a Cdk1 phosphorylation-dependent manner.

This report presents a novel regulatory mechanism involving the action of Cdk1/cyclin B1 and Plk1 upon their common substrate procaspase-8. Previously, we could demonstrate that procaspase-8 is phosphorylated in mitotic cells by Cdk1/cyclin B1 on S387, which is located at the N terminus of the catalytic subunit p10 (Matthess et al., 2010). Inhibition of Cdk1 activity by depletion of cyclin B1 or the use of the Cdk1 inhibitor RO-3306 enhanced the Fas-mediated activation and processing of procaspase-8 in mitotic cells suggesting the possibility that Cdk1/cyclin B1 alone exerts the anti-apoptotic effect in mitotic cells. In the light of our novel results we suggest a model that

describes S387 phosphorylated by Cdk1 as central part of a phospho-epitope that serves as a PBD-docking site for the subsequent phosphorylation of procaspase-8 S305 by Plk1. This cooperative mechanism imposes an additional layer of caspase-8 regulation by Plk1 to amplify the anti-apoptotic signal originally induced by Cdk1 activity. A recent study describes the sequential phosphorylation of protein tyrosine phosphatase 1B providing an additional example for the concerted action of Cdk1 and Plk1 to regulate cell death during mitosis (O'Donovan et al., 2013). A previous report describes the phosphorylation of procaspase-9 at T125 by Cdk1/cyclin B1 thereby regulating the threshold for activation of the intrinsic pathway during the cell cycle (Allan and Clarke, 2007). Future studies should address the question whether Cdk1/cyclin B1 might be the priming kinase for the sequential activity of Plk1 also in the case of procaspase-9.

Previous work by the Kim lab revealed a concerted action of Aurora A and Plk1 that results in a double phosphorylation of FADD (Jang et al., 2011a,b). A double mutant (FADD S194D/S203D) mimicking the phosphorylation of FADD by Aurora A and Plk1 induced cell death via activation of caspase-8 and caspase-3. Hence, the combinatorial action of Plk1 with additional kinases (Aurora A, Cdk1) in time and space seems to regulate the threshold for the extrinsic death pathway.

Cells, in which activation of caspase-8 alone suffices to stimulate procaspase-3 processing following death receptor-mediated apoptotic stimuli, are named Type I cells. In contrast, in Type II cells the intrinsic apoptotic pathway enhances caspase-8 mediated signaling and activation of caspase-3. Upon activation of caspase-8 in Type II cells, active BID (tBID) oligomerises the pro-apoptotic Bax protein, which in turn leads to the release of cytochrome-c from the mitochondria culminating in the activation of procaspase-9 (Eskes et al., 2000; Gillissen et al., 2013). Inhibiting Plk1 activity by BI 2536 resulted in a significant increase in overall apoptosis, upon Fas treatment, in the Bax knockout cells as compared to the Bax WT cells (Figure S2). Still, the knockout of the Bax gene alone is not sufficient to silence the intrinsic apoptotic pathway completely. Bak, another pro-apoptotic member of the Bcl-2 family, has similar and overlapping function in activating the intrinsic apoptotic pathway as Bax (Fulda and Debatin, 2006; Zhang and Fang, 2005). However, this further emphasizes that inhibiting Plk1-mediated phosphorylation of procaspase-8 at S305 can potentially sensitize even Type II cancer cells, with a muted intrinsic apoptotic pathway, to Fas-induced apoptosis. This is important, since a blocked intrinsic apoptotic pathway is considered to cause poor prognosis in cancer patients while imparting resistance to stimuli of the extrinsic apoptotic pathway or other chemotherapeutic agents that rely on inducing DNA damage to trigger apoptosis (Fulda et al., 2002; Zhang and Fang, 2005).

Taken together, the information from different signaling pathways that are integrated in diverse apoptotic signal transducers via phosphorylation might play an important role for the decision between cellular survival, apoptosis or necrosis. The cooperation of cell cycle kinases is likely to be important for the sensitivity of cancer cells arrested by anti-mitotic drugs toward drugs stimulating the extrinsic death pathway. Cocktails of small molecule inhibitors targeting Cdk1 and

Plk1 are likely to increase the vulnerability of cancer cells toward stimuli of death receptors.

5. Conclusion

Our results have demonstrated that Cdk1/Cyclin B1 and Plk1 act in concert for the post-translational modification of procaspase-8. The inhibition of Plk1, which acts downstream of Cdk1/Cyclin b1, is sufficient to sensitize cancer cells toward the Fas-mediated stimulation of the extrinsic death pathway. The pre-treatment of cancer patients with small molecule inhibitors targeting Plk1 should be tested in combination with stimuli of death receptors in future clinical trials as novel therapy option.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.molonc.2013.12.013>.

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